NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L22.2a Full-text search of the literature library + wind-engineering data extraction

Group First Lab (proving ground)

Status Promoted

Period 2026-10-13 18:16 — 2026-10-17 11:55

Stage conclusion

10-14 Promoted to the platform tool pages: all five acceptance groups passed, and Phase 2 non-first-page fact questions reached the target page on 22/27; Phase 3 blind test on 10-14: 21/30 reached the page (Chinese 9/10, Chinese-to-English 8/10, books 4/7, scanned books 0/3); books were weak, and after platform improvement 0062, round3 book page hits rose from 4/10 to 7/10 (meeting the target) while round4 remained 8/12; the desktop I drive is connected and inventoried: net new ~12.5k PDFs / 420k pages, with Phase 1 extracting 289k text-layer pages (1.8 GB on the workstation); desktop corpus 0069 went live (10-16), hitting the target page on 11/30 in the desktop-source blind test (vs. 24/30 for the two libraries), with improvement leads handed to the platform; 10-15 all tracks pushed forward: OCR of the remaining 105k pages completed at 10-16 02:38 (105,336 pages, handed to 0069), three-library metadata and work-level deduplication, 31,365 pages of djvu/pdg books handed to the platform (472 Chaoxing-format pages unresolved), and litwiki/knowledge trial results returned; ingestion awaits 0051 and C drive cleanup. 10-13 conclusion: retrieval only. Two libraries with 4,691 PDFs / 129k pages indexed locally by page; for non-first-page fact questions, top-5 results hit the exact page about half the time (English 8/14, Chinese originals 5/5, Chinese queries on English papers 3/5

after query translation, books 4/6), paper-level 1/27; platform integration underway (three phases, Phase 1 within the week), prototype frozen as baseline; `pages` Phase 1 is live (tested 10-13 22:40); OCR of 79 scanned documents / 5,422 pages handed to Phase 2. Profile extraction is Shelved (single-page text had 6/78 errors and 27/78 requiring qualification, with figure pages often mislocated); 0 CU; books on the other computer await information from Liang Hu

Project report

Full-Text Literature Repository Search + Wind Engineering Data Extraction

Objective

Answer "how to get the most value out of Liang Hu's paper and book repositories." Do not fine-tune large language models; test (1) full-repository full-text search with citable page numbers; and (2) extracting one category of wind engineering data from papers into structured tables. Task brief recorded.

Current Status

10-13 Approved for opening by Liang Hu (prompted by discussions at the Strata node). 10-13 18:20 Minimal trial dispatched to `ideas` (task brief copy): (A) build a page-level hybrid BM25 + small-embedding index across the entire repository, with pre-specified criteria on ⩾30 questions (top-5 hit on the correct paper + page ±1), compared against the existing card search; (B) run extraction on 30–40 downburst vertical profile papers (`z_max`, `U_max`, profile model parameters, `r/D`, page and figure numbers), verifying ⩾60 values against original pages to measure accuracy and verification labor. Downburst paper selection references the wind-field line list (`windfield_step2`, ~120 papers, read-only). Known baseline: Zotero holds 3,163 PDFs / 17.7 GB, of which 2,422 already have full-text caches; existing search covers only literature with built cards, with no full-repository full-text index. Books: on 10-13 the Lead located two sources on the workstation—87 books + 32 book chapters + 60 dissertations inside Zotero, plus the EndNote doctoral library (1,527 PDFs / 6.7 GB, including books such as Kaimal & Finnigan and Naudascher as well as Chinese

literature), now included in scope (supplement queued after the initial run within the same execution); additional books may reside elsewhere, pending word from Liang Hu.

Corpus Inventory (10-13 ideas audit)

PDF Attachments	In-Repo Deduplicated Candidates	Size	Text Layer / Whole-Doc No Text / Extraction Failed	Language (First-5-Page Heuristic: EN / ZH / Mixed / Unknown)	Type
3,164	2,592	16.5 GiB	3,061 / 83 / 20	2,959 / 2 / 95 / 108	73 books, 26 book chapters, 50 dissertations, 2,870 journal articles, 145 other (by attachment; item-level counts: 87/32/60)
1,527	1,348	6.2 GiB	1,440 / 76 / 11	1,402 / 0 / 35 / 90	1,517 unclassified (`.enl` metadata undecoded; 56 long documents ⩾150 pages)
4,691	~3,845 candidate work clusters (95 cross-repo same-title clusters, approximate)	22.8 GiB	128,708 pages indexed; 159 whole documents without text / 10,553 pages (not equal to total scanned count)	Very few Chinese documents have a text layer: Chinese books are mostly scans, pending OCR sampling	—

PDF Attachments	In-Repo Deduplicated Candidates	Size	Text Layer / Whole-Doc No Text / Extraction Failed	Language (First-5-Page Heuristic: EN / ZH / Mixed / Unknown)	Type
Literature 12,933 + Book 1,060 readable PDFs (plus 132 unreadable files); only 53 byte-level duplicates with the two existing repos → net addition of ~12,497 files / 421k pages (>3× the two existing repos); 62 `.djvu` books (~30k pages); 11 Chaoxing `.pdg` books (3,163 pages)	15.9 GiB net deduplicated	Sampling estimates ~127k pages lack a text layer and need OCR (~90% of Chinese-titled items in Book are scans), estimated at 80+ hours using 10 threads on the workstation	Sample of 668 items: 124 Chinese / 406 English or mixed / 138 unknown	Byte deduplication underestimates overlap (cannot recognize different versions of the same work); PhDBackup differs byte-wise from the EndNote doctoral library	

Settled Decisions

- Do not fine-tune; prioritize retrieval (Item 4 of the review verdict).
- No local large language model needed: retrieval uses small embedding models + keywords, while data extraction uses the batch Gemini channel.

Trials and Results

Date	Action Taken	One-Line Result
10-13	Supplement (`ideas`): incremental indexing of EndNote library + in-repo books, Chinese CJK trigram index + small glossary; added 6 book interior-page questions	English first-page accuracy unchanged after corpus expansion (hybrid 27/34). Only 1/6 book questions hit the page: English book questions 1/3 (B02 hit, B05/B06 missed), Chinese queries for English books 0/2, Chinese query for Chinese book B04 hit the book but missed the page; original 34 Chinese queries for English papers hit 0–2/34. Lead review: failures stem primarily from **cross-lingual mismatch** and **keyword-only first-stage recall** (the embedding model only reranks the top 60 keyword-recalled pages and understands only English), plus cross-repo duplicate copies of the same book occupying top-5 slots
10-13	Round 2 (`ideas`): 30 non-first-page fact questions (3 questions discarded as flawed, 27 valid), top 5 deduplicated by title + page; AI page-by-page verification of 78 extracted values; 8 papers with calibrated figure pages sent to Gemini for visual chart reading	Retrieval (top-5 paper + page): English facts 8/14 (paper found in 10–11/14); Chinese original text 5/5 (trigram index); Chinese queries for English text 0/5, rising to 3/5 after translating only the query; books 4/6 (physical books 2–3/4); baseline tool paper-level hit 1/27. Extraction: among 78 values, 45 supported by same-page text, 27 required qualifying conditions/definitions, 6 wrong; in 5 of 8 papers the target figure was actually on a different page; visual chart readings by AI and Gemini roughly agreed on 5 papers, error unassessed without human ground truth. Lead spot-checked ground-truth answers for 5 questions (R01/R07/R13/R21/R27), all confirmed on the original pages

Date	Action Taken	One-Line Result
10-13	Scanned PDF OCR (`ideas`): local Tesseract on 159 textless PDFs, 79 deduplicated files / 5,422 pages completed 22:00–22:35, 0 CU	5,407 pages yielded text; 20-page spot check found proper-noun/number errors on 8 pages and formula distortion on 8 pages; Lead checked against the original image on p. 40 of the Wilcox PDF: body text mostly accurate, formulas garbled. Usable for keyword recall, but cannot be cited directly for formulas or numerical values. Handed over to platform Phase 2 for ingestion
10-14	Platform acceptance (Phase 1 at 10-13 22:03 / Phase 2 at 10-14 01:58, per-question files)	Phase 1 passed all five groups: first-page 29/34, English facts 6/11 (valid-question basis; prototype 5/11), Chinese original 5/5, books 5/6, Chinese-to-English 3/5; Phase 2 `round2` hit 22/27 valid questions (target 20), English facts 9/11, 31/34 Chinese-to-English first-page questions hit the page, cross-repo duplicates merged. Weakness: the 6 Round 1 supplementary book questions hit the page in only 2/6 cases in both phases. Lead verified SHA prefixes and questions R13–R16 individually
10-14	Phase 3 blind test (`ideas`, 30 questions with frozen SHA, questions withheld from platform), single `pages` call per question	Page hit 21/30, work hit 25/30: Chinese original 9/10, Chinese-to-English 8/10, books with text layer 4/7, scanned books 0/3 (work hit 2/3). Book interior pages remain broadly weak (consistent with 2/6 on the Phase 1/2 book supplement); discovered metadata misattachment: OCR output for *Introduction to Tropical Cyclone Dynamics* displayed a 1949 paper title (verified by Lead in raw output). Aggregated results handed to platform for a dedicated Phase 3 improvement

Date	Action Taken	One-Line Result
10-14	Desktop Phase 1 (`ideas`): remote text extraction on 12,497 net new PDFs on the desktop, transferred back to the workstation over SSH without writing PDFs to disk	Succeeded on 12,489 files / 418,747 pages, including 288,794 pages with text and 129,953 textless pages (for Phase 2 OCR), plus 8 files with pagination anomalies (2,276 pages) pending diagnosis; actual workstation footprint 1.8 GB (small-file block overhead), C drive at 52 GB. Lead spot-checked 2 pages by re-extracting from the original PDFs; content matched
10-15	Desktop Phase 2 wrap-up (`ideas`): OCR halted at 24,615 pages (all Book/Chinese; auto-stopped at 10-14 20:46 when C drive dropped below 40 GB); exported package manifest; patched 2 anomalous PDFs	SQLite holds 418,861 page records / 12,491 ingestible files (394,132 text-layer + 24,615 OCR); 105,338 pages remain un-OCRed (~27k in other Book directories, ~69k in Literature), plus 6 corrupted PDFs (2,162 pages); 20-page spot check: frequent errors in formulas, numbers, and proper nouns. Handed over to platform for ingestion
10-15	Re-ran both blind test sets after platform book-retrieval update went live (`ideas`, `pages`)	`round3`: page hits rose from 21 to 24/30, books from 4/10 to 7/10 (meeting the Phase 3 target), Chinese unchanged; `round4` (12 book questions) unchanged at 8/12. Remaining failures are mainly whole books failing recall—once a book is recalled, the target page is almost always found; aggregated results handed to platform
10-15	`knowledge` bot trial of `pages` to verify sources on 10 knowledge cards	2 cards hit the exact source page, 2 hit related literature, 6 were not found. Lead checked Zotero: of the 6 unfound papers, 5 are absent from the repository and 1 has an entry without a PDF (`YY6ZFJJC`) → retrieval itself did not fail; the gap is that "users cannot tell whether an item was missed by search or is absent from the repo," so suggested the platform add a notice. Bot feedback: exact English title + specific numerical value/method works best; broad Chinese topics or bare DOIs get crowded out by neighboring papers

Date	Action Taken	One-Line Result
10-15	`LitWiki` trial of `pages` to verify 10 citations with known page numbers	First query hit the target page on 5/10, rising to 7/10 after rephrasing; all misses occurred when "the paper is already known and a specific page inside it is sought," where other papers in the same series or reference-list pages crowded out the result. `LitWiki` recommended integrating it into the card-building workflow for candidate page localization while still opening the original PDF to verify citations; suggested the platform add within-work filtering, exact-title boosting, and prominent labeling of inferred journal page numbers
10-15	Bibliographic metadata Round 1 (`ideas`, metadata workspace): decoded 1,373 entries from EndNote `.enl`, safely linked 1,122/1,527 attachments; all 12,491 desktop titles are low-confidence (7,092 from filenames, 5,399 from first-page heuristics; PDF Info unread due to expired socket); 15,009 candidate work clusters across the three repos	30-item spot check showed 23/30 usable: EndNote 9/10, desktop 5/10, cross-repo clusters 9/10. Usable only for candidate linking, not as a formal bibliography; only 34 desktop items shared a DOI or fingerprint with the two existing repos (deduplication remains a conservative lower bound). Round 2: (A) verify 912 DOI candidates against Crossref to backfill titles (dispatched); (B) wait for the re-extraction run to export `pdf_meta.jsonl`, then supplement PDF Info titles/authors and re-cluster

Date	Action Taken	One-Line Result
10-15	`.djvu`/`.pdg` text conversion (`ideas`, desktop working directory, completed at 14:25)	Without `sudo`, used `apt download` to unpack `tesseract` 4 + `djvulibre` into user space (viable and reused by other tasks). Delivered `djvu_pdg.sqlite`: 73 documents / 31,365 pages (61 DjVu books with 28,391 pages + Melenk recovered 282/283; 11 Chaoxing PDG sets with 2,692/3,163 pages), 31,140 pages with text (24,637 text-layer + 6,728 OCR); 472 failed page slots (398 PDG `0x38` transform, 72 auxiliary layers, concentrated in *Introduction to Wind Engineering* and *Steel Bridges*) awaiting a fuller decoder; 10-page spot check showed normal text. Lead verified workstation file SHA256 and page counts matched; handed over to platform for appending
10-15	Desktop OCR Phase 3 (`ideas`, working directory, background PID 63579 started at 10:29, 6 threads)	Reused unpacked Tesseract; 200-page trial took 36 s (~5.5 pages/s, 195 pages with text) → extrapolating 105,338 pages to ~5.3 hours, expected to finish afternoon/evening of 10-15; corrupted PDFs being repaired separately via `pypdf` copies. Upon completion, transfer `ocr_round2.sqlite` back to the workstation, export manifest, and submit to the platform
10-15	Desktop text re-extraction (`ideas`, working directory, 6 processes started at 10:47, PyMuPDF blocks sorted by left/right column)	Target: 12,437 files / 394,246 pages, carrying over OCR pages as-is; ~22 pages/s at 10:49; **completed at 13:2x** (12,436 files OK, 1 error, 394,243 text-layer pages + 24,615 OCR pages), wrap-up delivery and Bibliography B dispatched; concurrent execution slowed the OCR job to ~3 pages/s → ~10 hours (expected tonight). Desktop load ~14 (24 cores), still with headroom. Upon completion, deliver 50-page two-column comparison, manifest, and transfer back to the workstation

Date	Action Taken	One-Line Result
10-15	Bibliography Round 2A (`ideas`, sending only DOI strings to Crossref): 1,942 unique DOI candidates (desktop 912, EndNote 1,070)	1,178 source records matched Crossref titles (≥0.82 similarity) → backfilled high-confidence title/first author/year into `*_v2.csv`; 258 suspected false captures (DOI resolves but title mismatches); 22 non-existent; 522 DOIs containing OCR garble (e.g., Chinese characters mixed in) marked "unverified" (erratum: first edition mislabeled them as non-existent). **1,065 source records share a DOI with Zotero** (whereas byte deduplication found only 34) → true cross-repo overlap is far larger than previously estimated, so work-level merging must rely on DOI/title. 7 records with matching titles but year differences >1 year pending manual check
10-15	Re-extraction wrap-up + Bibliography Round 2B (`ideas`)	`v2` transferred back to the workstation (895 MB, 418,858 pages, SHA256 matched on both ends; 1 PDF yielded only 2 pages and is flagged anomalous); in the 50-page two-column sample, 17 pages confirmed continuous left-column sentences while the rest require manual inspection. Bibliography: 12,491 desktop items categorized into five title-confidence tiers (138 / 1,256 / 989 / 8,540 / 1,568); none of the 522 garbled DOIs could be safely repaired; 15,159 candidate work clusters: 12,272 desktop-only, 1,538 Zotero-only, 1,065 EndNote+Zotero (~70% of EndNote overlaps Zotero), 278 EndNote-only, and only 6 clusters overlapping between desktop and Zotero (recalculated by Lead; NUL bytes in CSV cleaned); 30-item spot check hit 24/30 (Crossref 10/10, PDF Info 6/10, EndNote 8/10). Handed over together with `v2` to the platform

Date	Action Taken	One-Line Result
10-16	Desktop OCR Phase 3 completed (`ideas`; interrupted at 10-15 23:3x when the desktop WSL root partition went emergency read-only, resumed at 10-16 00:49 after Liang Hu rebooted, delivered at 02:38)	105,336/105,338 pages (103,456 with text, 1,880 empty), 2 pages timed out during rendering, 6 corrupted PDFs excluded; `ocr_round2.sqlite` (256 MB) matched SHA256 on both ends with integrity OK (verified by Lead); 20-page spot check legible but frequent errors in Chinese glyphs, formulas, and tables. Handed over to the platform for appending. **Desktop corpus delivery now complete at ~556k pages**
10-16	Post-deployment acceptance (`ideas`): `round5` desktop blind-test initial run + `round3`/`round4` re-runs	`round3` reached the page on 24/30 and `round4` on 8/12, holding steady without regression (long-book recall improvements show no visible gain yet). `round5` initial run scored 5/30, but all 30/30 queries hit a translation channel outage (falling back entirely to Chinese retrieval), leaving both English-source groups at 0/10 and invalidating the run for a re-run (Lead verified at 20:58 that translation had recovered). Display checks: desktop source label and I-drive location were 8/8 correct, scanned-page notices 3/3; titles showed 4/8 as 'title candidate' and 3/8 as hexadecimal garble from PDF Info (reported to the platform for filtering)
10-16	`round5` re-run (translation recovered, 30/30 translated successfully)	**Desktop sources reached the page on 11/30 and the work on 15/30** (Book 4/10, papers 4/10, Chinese 1/5, OCR 2/5), noticeably weaker than the two existing repos (`round3` 24/30). The primary cause is failed work-level recall (15/30); the Chinese group actually dropped after translation succeeded (whereas the initial run using only raw Chinese queries hit 5/5 works) → Chinese queries should be searched along both Chinese and English routes; 12/15 titles still appear as 'title candidate'. All three improvement leads handed to the platform

Open Items and Next Steps

- **10-16 21:40**: Official `round5` score is 11/30 reaching the page (platform note: the tested build was the 09:42 version, which did not yet include the `ocr_round2` and `djvu_pdg` increments while embeddings were still being built). Four improvements (dual-route Chinese and English retrieval, separate recall from the desktop and the two existing repos followed by merging + title boosting, each copy using its own physical page numbers, and replacing filenames with `desktop_records_v3` titles) have been added to Round 3 of the platform update; regression requires no per-question drop on the two existing repos and a latency increase of ≤5 s. Once live, the platform will notify the Lab to re-test with the identical question set; no acceptance verdict on desktop sources will be drawn before then.
- **10-16 20:46**: `pages` is now returning desktop sources (the update is live, though the platform has not yet sent formal notice); dispatched `ideas` to run `round5` + `round3`/`round4` re-runs.
- **10-16 02:45 status**: All three desktop sources (`v2`, `djvu_pdg`, `ocr_round2`) handed over to the platform; awaiting ingestion and deployment → run `round5` 30-question blind test + longitudinal comparison against `round3`/`round4`. Remaining unresolved: 472 PDG pages (decoder), 6 corrupted PDFs, 2 timed-out pages.
- **10-15 09:05 Liang Hu: "Leave other directories on the I drive alone for now; push everything else forward."** Dispatched (with non-overlapping write scopes): ① `ideas` in-place OCR on desktop WSL for the remaining 105,338 pages (desktop has 24 cores / 46 GB RAM / 664 GB free disk; no `sudo`, switched to rootless `micromamba` + `conda-forge` to install `tesseract`, etc.; initial dispatch did not start due to expired socket, rebuilt 3-day socket at 09:20 and re-dispatched; output from working directory → workstation `ocr_round2.sqlite`); ② Bibliography metadata Round 1 delivered (candidate grade); Round 2A delivered (1,178 high-confidence backfills; 1,065 sharing a DOI with Zotero); Round 2B delivered (see Trials and Results); `v2` + bibliography tables handed to platform at 10-15 14:05, **cleared for ingestion at 14:1x** (new constraints: net addition ≤8 GB, halt if C drive drops below 85 GB; 117 GB free inside the virtual disk); old `page_text.sqlite` and old manifest deleted (14:20); `ocr_round2` and `djvu_pdg` to be appended upon delivery; `round5` 30 blind-test questions for desktop sources frozen (SHA recorded; 10 Book / 10 paper / 5 Chinese / 5 OCR), with the 5 OCR questions visually verified against original pages by the Lead, awaiting deployment for scoring; ③ `.djvu`/`.pdg` order delivered at 14:25 and submitted to the platform; OCR Phase 3 ran full-scale on the desktop starting at 10:29, **until at 23:3x (at 102,850/105,338 pages) the desktop WSL root partition entered emergency read-only mode (`ext4` error triggered automatic `remount-ro`) and the process exited**; backed up the SQLite file to the workstation temporary partial OCR backup directory (102,850 pages, integrity OK); after Liang Hu rebooted, resumed at 10-16 00:49 and completed delivery at 02:38 (see Trials and Results); temporary partial OCR backup `ocr_round2_partial` on the workstation deleted; re-extraction job also ran on the desktop starting at 10:47 (~5 hours); `ideas`

scheduled self-check-ins at 13:20 and 14:29, with a fallback check-in at 10-16 08:2x; ④ Platform reply (10-15): the two live repositories are already sorted by text blocks without column-mixing; desktop `page_text.sqlite` was extracted with `-layout` and must be re-extracted, as the ingestion update will only accept the re-extracted version → dispatched `ideas` on the desktop to re-extract with PyMuPDF blocks and deliver `page_text_v2.sqlite` (adding an `extractor` column); the update reserves `work_id` merging, title/author/year display, and multi-source incremental SQLite appending (primary key `sha256` + physical page, where later sources only fill empty pages or replace OCR pages). Start conditions: C drive ≥100 GB + delivery of re-extracted version; ⑤ `litwiki` and `knowledge` trials returned (see the two 10-15 rows in Trials and Results): retrieval itself did not fail, and two improvement suggestions (notice when item/PDF is absent from repo; `--key`/`--doi` within-work search) were submitted to the platform to be queued after the current ingestion update; ⑥ Queried the platform regarding incremental indexing and call volume. Untouched for now: other directories on the I drive; downburst extraction remains Shelved.

- 10-14 14:20 Phase 2 dispatched to `ideas`: local OCR on 129,953 pages (prioritizing Chinese Book items, wind engineering, codes/standards, etc.), measured at ~1 page/s on 6 threads (Chinese recognition is slow), ~1.5 days for the full batch; auto-stopped at 10-14 20:46 per rule when the C drive dropped below 40 GB, having recognized 24,615 pages (24,360 with text) and leaving ~105k pages; C drive fell to 24 GB at 22:24 (growth came from system runtime and temporary directories, not the Lab), urgently reported to the platform, and will not restart until the C drive recovers above 40 GB; 10-15 00:53 re-check: all A3 processes are gone, directory size is stable at 11 GB, C drive recovered to 21 GB. **Under Liang Hu's new 10-14 rule (workstation does not run compute or store compute data), a different location must be chosen before restarting the remaining ~105k pages of OCR** (running locally on the workstation requires platform approval; books cannot be uploaded to Colab; CRC group space is ~95% full), to be decided after C drive recovery (46 GB as of 10-15 02:32); proposed plan: run OCR in-place on the desktop WSL (data is already there, requiring no transfer and zero workstation footprint), subject to Liang Hu's approval to install Tesseract on the desktop; 10-15 02:45 Phase 2 interim deliverable (419k pages) handed to the platform for ingestion (to be dispatched once virtual disk compaction brings the C drive back above 100 GB); platform book improvements went live at 10-15 02:09, and the platform accepted the re-run results (Phase 3 target met); long-book work-level recall improvements merged into the desktop ingestion update (two-level book indexing by chapter/page-group, terminology boosting toward books, and ≥20 platform-built whole-book recall test questions), starting once the C drive exceeds 100 GB with the Lab running longitudinal comparisons once live; remaining 105k pages of OCR await Liang Hu's decision on installing Tesseract on the desktop; Phase 1 (419k pages) packaged with 1,003 pages verified character-by-character without error; 2 of 8 anomalous files had page numbers patched while `FE-Handbook` and others are corrupted; `.djvu`/`.pdg` cannot be converted on the workstation without installing software and are deferred; net workstation addition ≤4 GB; diagnose the 8 anomalous PDFs and evaluate `.djvu`/`.pdg`

along the way. Submit manifest to platform upon completion for ingestion after prerequisite tasks and C drive cleanup.

- **10-14 Promoted**: Search promoted into the platform tool `pages` (Phase 1 and Phase 2), available to all Leads and bots. English fact metric standardized to 11 valid questions (my earlier ⩾8/14 mislabeled the raw 14-question prototype score as the valid-question basis).
- Phase 3 blind test handed to platform (10-14 10:15): Chinese queries work well, book interior pages are weak, and an OCR book had a misattached title; platform opened a dedicated improvement item (within-book page localization for long books + OCR noise compensation, misattached title fix, cross-repo same-work detection, and dual-route Chinese translation fallback); once live, the platform will first run regression on the Phase 1/2 question sets and then notify the Lab to re-run the exact same blind test set and return per-question results; targets are book page hits substantially above 4/10, Chinese original ⩾9/10, and Chinese-to-English ⩾8/10. Original plan: the Lab holds the blind test questions (drafted and scored by `ideas`, submitted to Platform Lead before 10-15) without sharing the questions with the platform, merging into the platform's ~10-20 Phase 3 acceptance; focus on whether book interior pages are broadly weak, opening a dedicated Phase 3 improvement if so. Platform registered Phase 1 promotion on 10-14. Thereafter the prototype can be unfrozen.
- **Conclusion (10-13 20:55): Pursue retrieval only.** The retrieval component has status Pending promotion: its form is a platform tool rather than a research project; the proposal has been submitted to the Platform Lead for scheduling, with the Lab providing the prototype and regression question sets.
- **Data extraction component Shelved**: Why it does not work—profiles mostly reside in figures where every value must be paired with a figure number, radial position, and operating condition; single-page text extraction yielded 6/78 errors and 27/78 requiring added qualifications, while 5/8 papers mislocated the figure page altogether; expert paper-by-paper chart reading is currently required, so automation saves little labor. What is missing: reliable visual chart reading (with curve attribution) and human ground truth. When to restart: when a dataset need with a concrete downstream use arises (e.g., a research line requiring a downburst profile benchmark database) that can accept the labor cost of "initial AI extraction + expert paper-by-paper verification," or when multimodal chart-reading models achieve acceptable error against human ground truth on ⩾10 papers.
- OCR completed and handed to platform Phase 2 (see table above). 10-13 22:40 live test of Phase 1 `pages`: Chinese-to-English translation works, but cross-repo duplicate copies of the same book occupy top-5 slots—platform confirmed Phase 1 deduplication did not take effect (EndNote titles were truncated filenames); Phase 2 parses EndNote metadata + deduplicates by DOI / file hash / first-page fingerprint, and ingests OCR pages labeled "OCR text, verify original page before citing"; QuikSCAT query added as a regression case. Awaiting per-question Phase 1 acceptance results and Phase 2 acceptance results from the platform.

- 10-14 Connected to desktop WSL (Tailscale SSH, user account via IP; the hostname entry in `known_hosts` was stale, and the current host key matches Tailscale registration) and completed the inventory (see Corpus Inventory table); 10-14 Phase 1 dispatched to `ideas`: extract page-level text from PDFs with text layers, keeping net workstation addition ≤4 GB; defer OCR/`.pdg`/`.djvu` to Phase 2. Storage correction: the WSL disk is a `vhdx` on the C drive (355 GB), so the free space reported inside WSL is illusory and the true limit is the remaining C drive space (47 GB on 10-14); reported to platform. Platform reply (10-14): desktop text ingestion is queued after prerequisite tasks and awaits resolution of the C drive space issue (indexing 420k pages is projected to take 4–6 GB); results will be labeled "Source: Desktop" and display the original I-drive source location; the platform will open an ingestion item once the Lab submits the manifest. Reclaiming C drive space requires compacting the virtual disk with a 10–30 minute platform-wide downtime, now listed as Item 11 on the platform's decision list for Liang Hu. Previously: the workstation has no I drive; the Windows desktop host (online via Tailscale) rejected the workstation SSH key and had no SMB share enabled; the desktop reported a host key change, so we stopped without bypassing it. Requested Liang Hu to add the workstation public key or enable a read-only share; once accessible, perform an inventory first without copying files.
- Awaiting word from Liang Hu on the location, count, and scan status of books and papers on the other computer; destination for copies >5 GB to be decided by Liang Hu (leaning toward the local WSL disk).
- 10-13 Platform Lead decision to integrate: three phases, with Phase 1 within this week. Lab commitment: keep prototype scripts and question sets frozen until Phase 1 acceptance (communicated to `ideas`); regression question sets and passing thresholds returned to the platform (floor not below prototype: first-page ≥29/34; `round2` subgroups English ≥8/14, Chinese 5/5, books ≥4/6, Chinese-to-English ≥3/5; `round2` overall target ≥20/27). The platform recorded all five question sets and thresholds (10-13). Awaiting Phase 1 acceptance results, after which this page will be updated to Promoted; fallback check-in around the evening of 10-16.

Literature Basis

- Review verdict (Ovadia: unsupervised fine-tuning underperforms retrieval; Balaguer: supervised fine-tuning adds +6 pp).

Notes from Liang Hu

- 10-13: "This doesn't require a local large language model, right?" — Right.
- 10-16 02:4x OCR Phase 3 (scanned pages among the 12,491 desktop items) completed: 105,336 pages across 1,713 files, with 2 rendering timeout pages excluded (FORTRAN90

manual), integrity OK, SHA256 verified; during the run the desktop root partition hit a read-only fault and the SQLite copy on the local mounted drive (`9p`) was corrupted (lesson: never run SQLite writes on a Windows-mounted drive). Handed over to the platform for ingestion.

Key figures

than the ABL–downburst case. However, at $\alpha = 180^\circ$ the temporal shift is smaller because the ABL and downburst flow align and the downburst front propagates more rapidly. The full-scale data remain in good agreement with both laboratory measurements and CFD results.

A comparison in terms of turbulence energy spectra between experimental laboratory data and LES data is presented in Fig. 9 for $\alpha = 0^\circ$ at $R/D = 1.0$ and $\alpha = 180^\circ$ at $R/D = 1.4$, respectively. The spectra are computed from the rapidly-varying reduced turbulent fluctuations ($\widetilde{u}$), which are derived based on the wind velocity decomposition technique for non-stationary phenomena (Chen and Letchford, 2004; Solari et al., 2015). Hereby, the moving mean time window $\Delta T = 0.1$ s is firstly used to evaluate the slowly varying

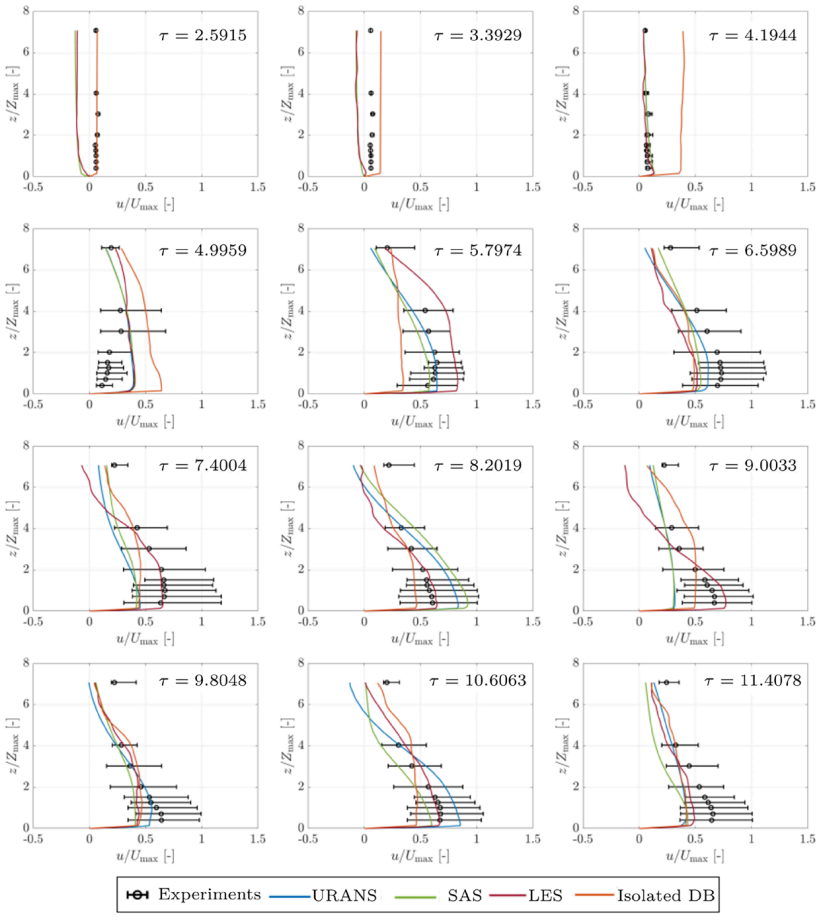


Fig. 11. Comparison of experimental and simulated vertical profiles of radial velocity (u) for α = 0° at R/D = 1.0, at a selected set of non-dimensional time steps (τ) (see also Fig. 2). The result of the isolated downburst (DB) wind simulated by Žužul et al. (2024) is also shown for comparison.

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3. Flow field

Mason, *et al.* (2005) reported details of the simulated downburst wind field produced using the pulsed jet simulator. However, to aid in the understanding of the loading results presented in this paper, a brief discussion of the flow field is described.

The physical structure of the wind field impacting the model cube is made up of two parts. The first part being the initial non-stationary flow due to the pulsing of the jet, or in atmospheric terms, the gust front, and the second is the stationary field due to the steady impinging jet. The non-stationary component is made up of a succession of vortices with intermediate turbulent flow between them. The model cube first experiences what is termed the primary vortex, which is produced due to the flow instability occurring at aperture opening. Passage of this vortex is followed by an intermediate flow region that precedes the passage of a larger trailing vortex. The latter vortex was seen to be due to a combination of aperture effects and the natural shedding frequency of the jet. The trailing vortex is produced primarily as a side effect of the experimental setup (i.e., shedding); it is not known to what extent this trailing vortex would exist in a true downburst event. After the non-stationary component of flow, the cube experiences a succession of smaller vortices fed by the shear layer shed from the jet outlet. This flow characteristic has previously been documented in steady impinging wall jet flow (see for example Popiel 1991, Zaman and Husain 1980). In this paper however, it is the non-stationary region of the flow that is of interest.

The most important characteristics of the wind field for this particular experiment are the vertical (z) profiles of the surface parallel velocity, Fig. 3. For each elevation the peak velocity corresponding to the passage of the primary vortex was averaged over the five individual test runs to provide the profile shown. Mean velocities associated with the steady flow region are also shown. Velocities are shown normalised against the mean jet centre line velocity, V_j, while elevations are normalised against the jet diameter, D. The peak values represent something of an envelope of velocities associated with the vortex, and do not represent instantaneous structure. Steady flow profiles are shown as an

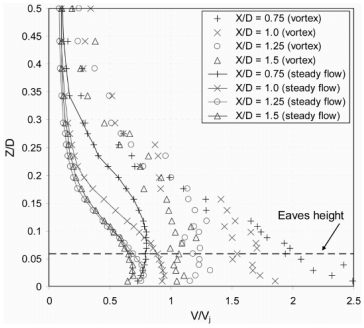


Fig. 3 Steady flow mean, and ensemble-averaged peak velocity (within the vortex) profiles at selected radial locations

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(accuracy ±1mm representing a percentage error of ±0.04%). Vertical distance measurements were undertaken with steel rulers to which Vernier gauges were fitted, enabling measurements of the order of ±0.1mm to be made. The majority of the data presented below were measured at vertical distances of approximately 20mm above the ground, resulting in a percentage error of the order of 0.5%.

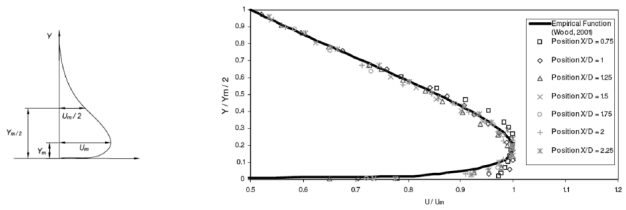
3.4. Ensemble averaging

It was observed early in the experimental programme that, whilst the measured velocities were well defined when a mean flow was established, there was significant run-to-run variation of the experimental results for the developing flow field around the gust front, resulting from the large turbulence scales generated in the flow field. Thus, unless indicated otherwise, results presented below that describe the transient effects are presented as "ensemble averages", where 10 time histories for any particular measurement case are aligned and the results averaged at each time to give an ensemble mean and an ensemble standard deviation. In order to enable comparison between the different datasets, the time is assumed to be zero when the velocity corresponds to 40% of the peak velocity, although this must be regarded as an arbitrary definition.

3.5. Preliminary experiments

Constant outflow velocity

A series of preliminary experiments was undertaken for a constant outflow velocity to investigate the nature of the velocity field within the downdraft and the effect of nozzle height above the ground plane. Normalised velocity profiles are shown in Fig. 4, with the notation adopted for the normalization shown in Fig. 4a. Fig. 4b illustrates the results of the normalised streamwise velocity profiles obtained at various normalized distances (X/D) from the stagnation point for a nozzle height of 2D. In Fig. 4b the vertical axis represents the vertical height (Y) above the ground normalized by the height corresponding to half of the streamwise maximum velocity ($Y_{m/2}$). This normalization is common in terms of representing the flow field associated with a wall jet (see Wygnanski, *et al*. 1992). Also shown in Fig. 4b is the empirical function of Wood, *et al*. (2001) illustrating agreement of the current data with previous studies for values of $Y/Y_{m/2} > 0.2$. It is also



(a) Normalization notation. (b) Experimental data.

Fig. 4 Normalised streamwise velocity profiles under steady state conditions, H/D = 2.0

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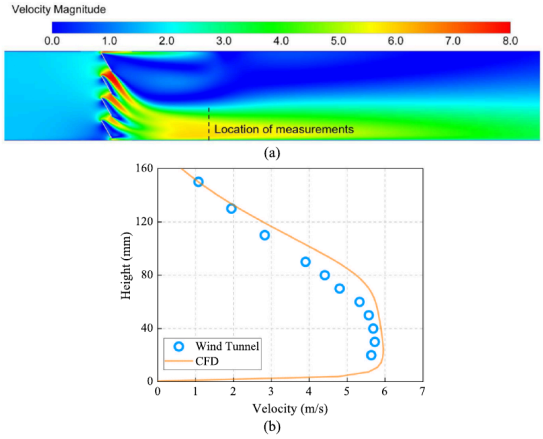


Fig. 15. Simulation of downburst outflow based on CFD with blade rotation angle of 60°: (a) CFD velocity contour; (b) comparison of vertical profiles between wind tunnel experiments and CFD simulations.

noted that the blades could be furtherly adjusted for the requirements of wind tunnel experiments with arbitrary length scales.

Fig. 18 presents the velocity vectors corresponding to two scales discussed in Fig. 17. The arrows in each subfigure are scaled and colored according to the wind velocity. Fig. 18(a) demonstrates the presence of a primary vortex at the upper of the test chamber due to the larger rotation of Blade 1 and Blade 2, directing more intense downburst airflow. This observation results in the peak of the nose-like wind velocity profiles at the elevation of 37 mm. However, a smaller primary vortex appears in Fig. 18(b) due to the slighter rotation of Blade 1 and Blade 2, and multiple flow structures converge at higher elevations. With the contribution of the anticlockwise rotation of Blade 4, the maximum velocity occurs at the elevation of 85 mm. Besides, Fig. 18 also presents the angle of attack profiles. As shown in Fig. 18(a), the angle of attack increases with height due to the significantly redirected flow, but maximum angle of attack is less than 2 deg. Similarly, the

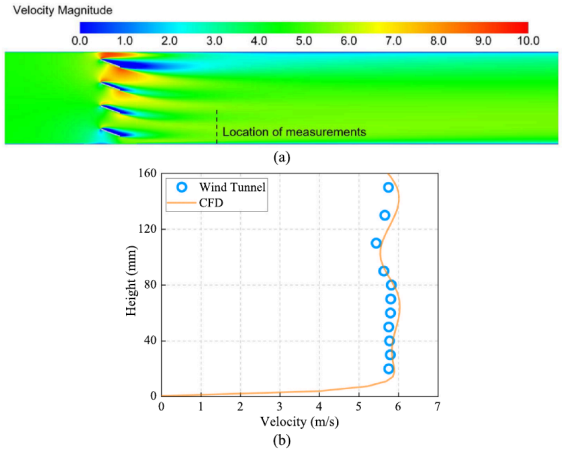


Fig. 16. Simulation of downburst outflow based on CFD with blade rotation angle of 20°: (a) CFD velocity contour; (b) comparison of vertical profiles between wind tunnel experiments and CFD simulations.

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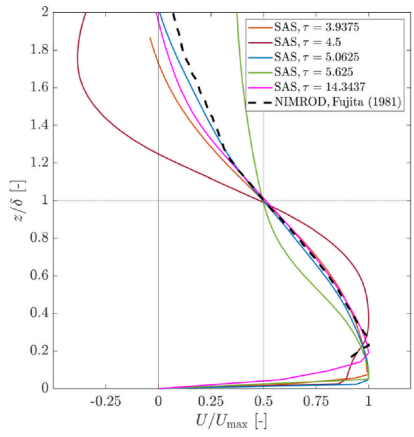


Fig. 15. Comparison of vertical profiles of radial velocity for dimensionless timesteps (τ) with the field measurements from the NIMROD project (Fujita, 1981).

- the flow within the bell-mouth is dominated by continuous unsteady separation of vortices from the high curvature of the nozzle, causing the interaction with the just-developed ring vortices.

4.3. Heights of the maximum radial velocity

Downburst winds can cause high-velocity winds in the near surface region, with potential to damage low-rise and mid-rise structures. Therefore, it is of great interest to analyze at which heights (z_{max}) radial velocities have their maximum values recorded throughout the downburst life cycle (*i.e.* the heights at which vertical profiles exhibit a peak as seen in Fig. 10). Fig. 20 shows the time history of heights at which the maximum radial velocities are recorded for $R/D = 1.2$. The *x*-axis shows the normalized time (τ), while the *y*-axis shows z_{max} normalized by the height at which the overall radial velocity maximum was observed in the experimental campaign (Z_{max}). Hereby, the experimental data is presented in terms of two-sided error bars centered around the average value of z_{max} observed across all twenty individual repetitions. The error bars hence represent the maximum and minimum observed height among the repetitions. The first parts of the signals (*i.e.* $\tau < 4$) can be disregarded from this analysis since the PV did not yet develop over $R/D = 1.2$ (resulting in low velocities). Generally, the measured mean z_{max} shows approximately constant values, but that is simply due to the high content of fluctuations (*i.e.* positive and negative values) contained within experimental repetitions which smooth the signal. Slight general trend of increasing measured mean of z_{max} is observed after the ring vortex passage ($\tau > 5$) that indicates the underlying tendency of individual repetitions of recording higher values of z_{max}. Conversely, both SAS and URANS data exhibit an oscillatory behavior with respect to measured mean data. These oscillations are generally associated with the passage of ring vortices. Moreover, prior and during the passage of ring vortices, z_{max} occurs at lower locations, while it is shifted upwards after the vortex passage. Overall, the height z_{max} at which the maximum radial velocity is recorded at particular R/D location varies throughout the downburst life cycle as a function of the passages of ring vortices.

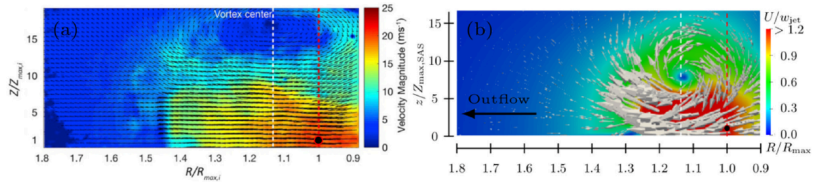


Fig. 16. Qualitative comparison of flow field during the maximum radial velocity occurrence between the SAS simulation and the PIV measurements by Junayed et al. (2019).

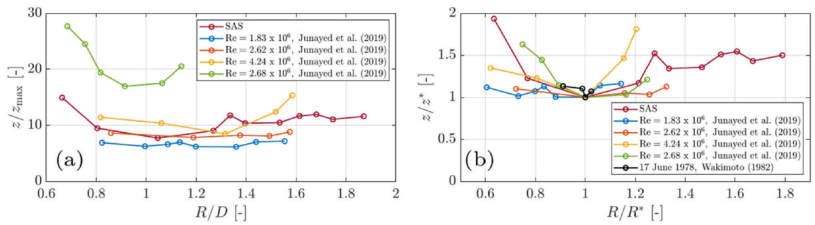


Fig. 17. Comparison in terms of vortex center locations of: (a) SAS and PIV measurements by Junayed et al. (2019), (b) SAS, PIV measurements of Junayed et al. (2019) and full-scale downburst event of 17 June 1978 by Wakimoto (1982).

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AI-generated, not peer reviewed; exploratory trial, not a formal project closure.